Gabriel Røer

Robotics student | Aerospace Technologist

GabrielRoer@icloud.com · +47 469 10 340 · Ås - NMBU · VikingInOrbit · gabriel-røer-0233713a4

PROFILE

Qualified space technologist and third-year Master's student in Applied Robotics at NMBU, with practical experience in embedded systems, PCB design, programming, systems engineering, aerospace and technical lighting. Leader of Eclipse NMBU and former team leader of the winning team in the 2022 Norwegian CanSat Competition. Combines technical work with extensive experience in organisational leadership and youth work.

EXPERIENCE

Leader (CEO) | Eclipse NMBU — Oct 2026 ~present

- Elected leader by the general assembly of Eclipse NMBU, the student space technology association.

Youth politics and organisational work | Ungdomsråd, Ungdom og Fritid og elevdemokrati — 2014 ~nå

- Around ten years of youth participation and organisational work, from youth and student councils to county and national work in Ungdom og Fritid.
- Worked to ensure that children and young people's views are heard and that young people have real influence in matters affecting them.

Permanent employee (20%) | Kjeller'n Fritidssenter, Ski — Aug 2019 ~present

- Facilitates and follows up children and young people aged 10 to 25 at the club and in schools.
- Technical support at events and other activities.
- Started as a user at Kjeller'n in 2013, joined the youth crew from 2017 to 2019 and completed two youth leadership-development seminars.
- Represented Kjeller'n on Ski municipality's children and youth council from 2017 to 2019.

Project Lead | Eclipse NMBU — Nov 2025 ~Sep 2026

- Fenrir, from 27 November 2025: Led the development of the circuit board for Eclipse NMBU's first probe.
- studentraketten, from 29 May 2026: Project lead for the association's new student rocket.
- Atlas2, from 5 August 2026: Project lead for version two of the probe satellite.

National service, aircraft soldier | Air Defence Battalion, Forsvaret (Ørland and Evenes) — Aug 2022 ~Aug 2023

- Assessed as hardworking, considerate and dedicated, and fit for further service.
- Courses: First Responder; basic military driver training; MB 240GD/290GD/300GD; MB field vehicle module H; and tail-lift training for heavy vehicles.

Work-experience placement | Kirkebygden barne- og ungdomsskole, Enebakk — 4–8 Jun 2018

- Had no absence during the work-experience week and delivered full teaching sessions at the school.
- Praised for spotting tasks without being asked, guiding pupils without giving answers, handling conflicts and delivering a successful lesson on Sudoku.

EDUCATION

Five-year Master's in Applied Robotics | Norwegian University of Life Sciences (NMBU), Ås - NMBU — 2023 ~2028

Selected courses: Project in advanced programming, applied machine learning, control engineering, modern workshop technology and automation.

Vg3 Space Technology | AVS (Andøy videregående skole), Andenes — 2021 ~2022

Qualified space technologist with vocational competence and general university admission certification. Subjects: electronics and communication technology, space and satellite technology, telemetry, space physics, remote sensing and GIS, and Physics 1.

Vg2 Data and Electronics | Ås videregående skole — 2020 ~2021

Subjects: data and electronic systems and electronic infrastructure.

Vg1 Electrical and Computer Technology | Ås videregående skole, Ås — 2019 ~2020

Subjects: data and electronic systems, electrical energy systems, automation systems and vocational specialisation.

LEADERSHIP AND ORGANISATIONS

Leader (CEO) | Eclipse NMBU — 30. sep. 2026 ~nå

Elected by the general assembly and responsible for leading the student space-technology association.

Chair (adult) | Ungdom og Fritid Akershus — 2023 ~present

Member, Lighting Committee | Samfunnet på Ås - NMBU — 2023 ~nå

Provides technical lighting for live events and handles time pressure and unexpected situations calmly and systematically.

Project Lead and Component Lead | Eclipse NMBU — nov. 2025 ~sep. 2026

- **studentraketten** — 29. mai 2026 ~nå
  Project lead for studentraketten.
- **Atlas2** — 5. aug. 2026 ~nå
  Project lead for Atlas2.
- **Fenrir** — 27. nov. 2025 ~sep. 2026
  Component lead for Fenrir.

Founder and team leader, Team Trident | CanSat, AVS (Andøy videregående skole), Andenes — 2021 ~2022

- Led a team of six students to victory in the Norwegian CanSat Competition 2022 and participation in the European final.
- Coordinated a small team, assigned work according to members' strengths, and followed up details and solutions throughout the project.

Deputy leader and student representative, student council | AVS (Andøy videregående skole), Andenes Aug 2021 ~Jun 2022
Represented students on the council and board and chaired meetings in the leader's absence.

Leader, Nordre Follo Youth Council | Nordre Follo kommune 2014 ~2021
Responsible for engaging members, structuring the council's work and contributing to youth participation in the municipality. Described by the mayor as a confident, inclusive and unifying leader.

Co-organiser and facilitator, participation workshop | Nordre Follo kommune/ Reell medvirkning 20 and 23 Oct 2020
Participated in training and delivery of a workshop on children and young people's voices and participation in the municipality, for 50 young people over a total of 10 hours.

Student council representative | Ski ungdomsskole, Ski 2017 ~2019
Elected student representative. Participated actively in the student council, argued for viewpoints and considered others' opinions during disagreements.

Chair, interim county board | Ungdom og Fritid Viken 2019
Led the establishment of the county organisation and worked to set up the board's first meeting.

PROJECTS

Combined independent technical portfolio | Independent projects Ongoing

- Develops reusable Python and C tools, including GabesPythonToolBox and GabesEmbededToolBox, for state machines, configuration, filtering, logging, serialisation, I2C and telemetry.
- Builds automation and technical solutions with Raspberry Pi, Home Assistant and addressable lighting, and programs stage lighting on grandMA3.
- Explores outdoor robotics, media and knowledge workflows, and interdisciplinary systems for robotics and space technology.

Governance and document system | Eclipse NMBU 2026

- Produced case papers, ballots, responsibility letters, protocol structure and a plain-language guide so that the general assembly could run as smoothly as possible.
- Built the documents in LaTeX with document IDs, templates and cross-references for easy navigation by members.

Fenrir firmware | Eclipse NMBU 2025 ~2026

- Worked on bare-metal firmware for STM32U5G9xx, including boot orchestration, fault handling, clock tree, RTC, IWDG and register-level GPIO.
- Documented sensors and interfaces for IMUs, magnetometer, pressure and environmental sensors, power monitoring and logging.
- Produced a flight OS for Fenrir from bare-metal components to support operational reliability and system understanding.

AWARDS

Armed Forces medal for completed service | Forsvaret 2023

Winner, CanSat Norge | Team Trident 2022

Ildsjelprisen | Nordre Follo kommune 2020

SKILLS

Programming: Python, C, C#, embedded C, SQL, LaTeX
Embedded systems: STM32U5, Raspberry Pi CM4, ATTINY85, PCB design and routing in KiCad, sensors, SPI, I2C and SDMMC
Controls and estimation: PID, EKF/AHRS, stability, state space and sensor fusion
Systems engineering: Requirements specifications, product breakdown structures, knowledge organisation and risk assessment
Tools: Git, GitHub, LaTeX, Docker, Raspberry Pi, Home Assistant, grandMA3, Obsidian
Languages: Norwegian and English, both at working level

REFERENCES

Stein Nøsting, leader of the youth clubs in Nordre Follo kommune (phone: +47 416 01 477)

Karen Malene Kleiven Eide, former leader, Eclipse NMBU (attest 3 October 2026)

Hanne Opdan, mayor, Nordre Follo kommune (attest 18 August 2021)
Attests and diplomas available on request.

ATTACHMENT LIST